

## Worksheet 3: Atmospheres: Structure, Clouds, & Climate

### Full solutions

Covers Lectures 5 and 6: vertical structure, radiative balance, clouds, winds, climate

Planetary Systems (WBAS002-05) • Kapteyn Astronomical Institute, University of Groningen

This worksheet is ungraded practice, with the same shape and level as the final exam. Plan up to 2 hours for a serious pass. Work each problem through on paper before you open the solutions, and show your algebra: the exam asks for the steps, not only the number. Some parts state an intermediate result (“show that...”); use it to check your work, and to continue if a step resists. Carry at least four significant figures through the intermediate steps (the worked solutions print five) and round only at the end. All parts are analytical; no computer is needed.

#### Constants and data

$k_B = 1.381 \times 10^{-23} \text{ J K}^{-1}$     $m_u = 1.661 \times 10^{-27} \text{ kg}$     $\sigma = 5.670 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$     $S_\oplus = 1361 \text{ W m}^{-2}$  (solar constant at 1 AU)    $M_\oplus = 5.972 \times 10^{24} \text{ kg}$

Earth:  $P_0 = 1.013 \times 10^5 \text{ Pa}$ ,  $R_\oplus = 6371 \text{ km}$ ,  $T_{\text{eq}} = 255 \text{ K}$  (the equilibrium temperature of Lecture 5), surface  $288 \text{ K}$ ,  $\Gamma_d = 9.8 \text{ K km}^{-1}$

Water:  $L_v = 2.50 \times 10^6 \text{ J kg}^{-1}$ ,  $R_v = 462 \text{ J kg}^{-1} \text{ K}^{-1}$  (use  $L_v/R_v = 5400 \text{ K}$  throughout), triple point ( $273 \text{ K}$ ,  $611 \text{ Pa}$ )

| Body            | $T \text{ (K)}$ | $\mu$ | $g \text{ (m s}^{-2}\text{)}$ | Bond albedo $A$ |
|-----------------|-----------------|-------|-------------------------------|-----------------|
| Venus (0.72 AU) | 737 (surface)   | 43.4  | 8.87                          | 0.77            |
| Earth (1 AU)    | 288 (surface)   | 28.97 | 9.81                          | 0.30            |
| Titan           | 94 (surface)    | 27.4  | 1.35                          | —               |

Atmospheric data as in the Lecture 5 notes;  $\mu$  is dimensionless, in units of  $m_u$ . The young Sun had  $L/L_\odot = 0.71$  (Lecture 6 notes).

### Problem 1 How thick is an atmosphere?

The scale height sets the thickness of every atmosphere in the solar system, and hydrostatic balance connects the surface pressure to the mass sitting overhead.

(a) Compute the scale height  $H = k_B T / (\mu m_u g)$  for Earth and for Titan.

**Solution.** For Earth, with  $T = 288 \text{ K}$ ,  $\mu = 28.97$ ,  $g = 9.81 \text{ m s}^{-2}$ :

$$H_\oplus = \frac{k_B T}{\mu m_u g} = \frac{1.381 \times 10^{-23} \times 288}{28.97 \times 1.661 \times 10^{-27} \times 9.81} = \frac{3.9773 \times 10^{-21}}{4.7205 \times 10^{-25}} = 8425.6 \text{ m} \approx 8.426 \text{ km}.$$

For Titan, with  $T = 94 \text{ K}$ ,  $\mu = 27.4$ ,  $g = 1.35 \text{ m s}^{-2}$ :

$$H_{\text{Titan}} = \frac{1.381 \times 10^{-23} \times 94}{27.4 \times 1.661 \times 10^{-27} \times 1.35} = \frac{1.2981 \times 10^{-21}}{6.1440 \times 10^{-26}} = 21128 \text{ m} \approx 21.13 \text{ km},$$

matching the table in the Lecture 5 notes (8.4 and 21 km).

**|**  $H_{\oplus} \approx 8.426 \text{ km}$  and  $H_{\text{Titan}} \approx 21.13 \text{ km}$ , **matching the 8.4 and 21 km in the Lecture 5 notes.**

**(b)** Titan is far colder than Earth, and its nitrogen atmosphere has nearly the same  $\mu$  as air. Explain, from the structure of the scale-height formula, why Titan's atmosphere is nonetheless more than twice as extended as Earth's.

**Solution.** The scale height compares thermal energy  $k_B T$  against the gravitational confinement  $\mu m_u g$ . Titan's temperature is lower by a factor of about 3 (94 vs 288 K), which by itself would compress the atmosphere. But Titan's surface gravity is weaker by a factor of about 7 (1.35 vs  $9.81 \text{ m s}^{-2}$ ), and  $H \propto T/g$  with  $\mu$  essentially equal for the two nitrogen atmospheres. The gravity ratio wins:  $(288/94) \approx 3.1$  against  $(9.81/1.35) \approx 7.3$ , so  $H_{\text{Titan}}/H_{\oplus} \approx 7.3/3.1 \approx 2.4$ . Weak confinement, not warmth, is what puffs up Titan's atmosphere.

**|**  $H \propto T/g$ : **the gravity ratio** ( $9.81/1.35 \approx 7.3$ ) **dominates over the temperature ratio** ( $288/94 \approx 3.1$ ), **so**  $H_{\text{Titan}}/H_{\oplus} \approx 7.3/3.1 \approx 2.4$ . **Weak confinement, not warmth, puffs up Titan's atmosphere.**

**(c)** Integrate hydrostatic balance  $dP = -\rho g dz$  over the whole column to show that the surface pressure equals the weight per unit area of the overlying gas,  $P_0 = m_{\text{col}} g$ , where  $m_{\text{col}}$  is the column mass per unit area. Evaluate  $m_{\text{col}}$  for Earth, and from it the total mass of Earth's atmosphere.

**Solution.** Integrating from the surface to the top of the atmosphere, where  $P \rightarrow 0$ :

$$P_0 - 0 = \int_0^{\infty} \rho g dz = g \int_0^{\infty} \rho dz = g m_{\text{col}},$$

with  $g$  taken constant over the thin atmospheric shell. Therefore

$$m_{\text{col}} = \frac{P_0}{g} = \frac{1.013 \times 10^5}{9.81} = 1.0326 \times 10^4 \text{ kg m}^{-2},$$

about ten tonnes of air on every square metre. The total atmospheric mass is the column mass times the surface area:

$$M_{\text{atm}} = 4\pi R_{\oplus}^2 m_{\text{col}} = 4\pi (6.371 \times 10^6)^2 \times 1.0326 \times 10^4 = 5.267 \times 10^{18} \text{ kg},$$

about one millionth of Earth's mass.

**|**  $m_{\text{col}} = 1.0326 \times 10^4 \text{ kg m}^{-2}$  and  $M_{\text{atm}} = 5.267 \times 10^{18} \text{ kg}$ , **about one millionth of Earth's mass.**

## Problem 2 Sunlight in, infrared out

Radiative equilibrium sets the temperature every atmosphere starts from; the greenhouse

effect explains why surfaces are warmer than that.

**(a)** Compute the solar flux at Venus's orbit, then the equilibrium temperature of Venus,

$$T_{\text{eq}} = \left[ \frac{(1 - A) S}{4\sigma} \right]^{1/4},$$

and compare it with the value in the Lecture 5 notes.

**Solution.** The flux falls off with the inverse square of distance:

$$S_{\text{Venus}} = \frac{S_{\oplus}}{(0.72)^2} = \frac{1361}{0.5184} = 2625.4 \text{ W m}^{-2}.$$

With  $A = 0.77$ :

$$T_{\text{eq}} = \left[ \frac{0.23 \times 2625.4}{4 \times 5.670 \times 10^{-8}} \right]^{1/4} = [2.6624 \times 10^9 \text{ K}^4]^{1/4} = 227.2 \text{ K},$$

matching the notes' 227 K. Venus's brilliant clouds reflect so much sunlight that its equilibrium temperature is lower than Earth's 255 K, despite Venus being closer to the Sun.

**|**  $S_{\text{Venus}} = 2625.4 \text{ W m}^{-2}$  and  $T_{\text{eq}} = 227.2 \text{ K}$ , **matching the notes' 227 K: Venus reflects so much sunlight that its equilibrium temperature is lower than Earth's 255 K.**

**(b)** Venus's measured surface temperature is 737 K, more than 500 K above the value you just computed. Explain the mechanism that maintains this difference. Your answer must name the quantity that the atmosphere blocks, and why blocking it raises the surface temperature.

**Solution.** The 92-bar  $\text{CO}_2$  atmosphere is nearly transparent to incoming sunlight but nearly opaque to the outgoing thermal infrared radiated by the surface. The surface can only shed energy through an atmosphere that absorbs and re-emits infrared many times over, part of it back downward. To push the same absorbed solar power out through this blanket, the surface must radiate far more strongly, which by the Stefan-Boltzmann law means it must be far hotter. The equilibrium temperature still describes what the planet radiates to space, but that radiation now leaves from cold, high atmospheric layers rather than from the ground.

**| The 92-bar  $\text{CO}_2$  atmosphere blocks outgoing thermal infrared from the surface, forcing the ground to radiate far more strongly and heat up to push the absorbed solar flux to space.**

**(c)** A fellow student writes: "back-radiation from the atmosphere is a second energy source, so the surface receives more energy than the Sun provides, and that violates energy conservation." Judge this claim and correct what is wrong with it.

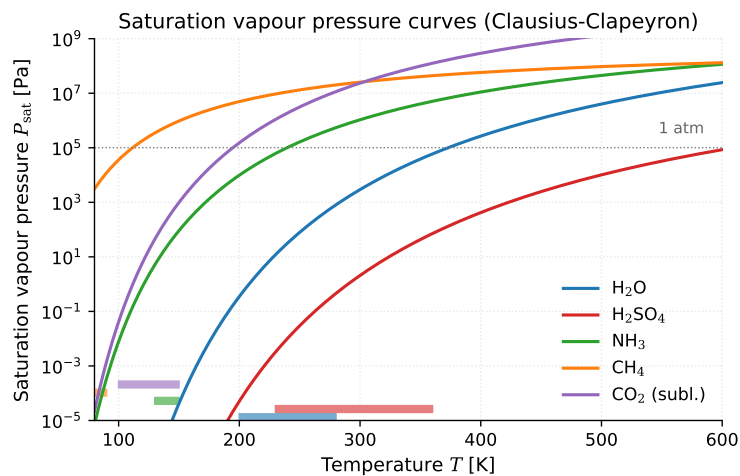
**Solution.** The observation is right and the conclusion is wrong. The surface does receive more radiative energy than the direct sunshine alone (on Earth, the downward infrared from the atmosphere is even larger than the absorbed solar flux), but the atmosphere is not a source: it only returns part of the energy the surface itself radiated upward. Every joule still originates from the Sun, and at the top of the atmosphere absorbed sunlight and outgoing infrared balance exactly. Energy is conserved at every level; the greenhouse effect only slows the escape of energy and raises the

temperature the surface needs to pass its share outward.

**False: the atmosphere is not a new energy source but recycles energy originally radiated by the surface, conserving energy at every level while slowing its escape to space.**

### Problem 3 From vapour to cloud base

The Clausius-Clapeyron equation from the blackboard derivation of Lecture 6 predicts where clouds form. The figure below reproduces the saturation curves from the notes; part (b) reads it.



**(a)** A parcel at the surface has  $T = 293$  K and relative humidity 50%. Use the integrated Clausius-Clapeyron equation with the triple point of water as the reference state to compute its vapour pressure (show that  $P_{\text{sat}}(293 \text{ K}) \approx 2357$  Pa and  $P_{\text{vap}} \approx 1179$  Pa), invert the Clausius-Clapeyron equation for the dew point (show that  $T_d \approx 282.4$  K), and from the dry adiabatic cooling rate find the altitude of the cloud base. Take the vapour pressure as constant during ascent.

**Solution.** First the saturation pressure at the surface temperature:

$$P_{\text{sat}}(293) = 611 \exp\left[-5400\left(\frac{1}{293} - \frac{1}{273}\right)\right] = 611 \exp(1.3502) = 2357 \text{ Pa},$$

so at 50% relative humidity  $P_{\text{vap}} = 0.5 \times 2357 = 1179$  Pa (checkpoints match). The dew point is the temperature at which this vapour pressure saturates. Inverting the Clausius-Clapeyron equation:

$$\frac{1}{T_d} = \frac{1}{273} - \frac{\ln(P_{\text{vap}}/611)}{5400} = 3.6630 \times 10^{-3} - \frac{\ln(1.9296)}{5400} = 3.5413 \times 10^{-3} \text{ K}^{-1},$$

so  $T_d = 282.4$  K, matching the checkpoint. Rising air cools at  $\Gamma_d = 9.8 \text{ K km}^{-1}$  while its vapour pressure is taken as constant (as instructed; in reality it falls slowly with the total pressure), so condensation begins where the parcel temperature meets the dew point:

$$z_{\text{LCL}} = \frac{T - T_d}{\Gamma_d} = \frac{293 - 282.4}{9.8} = 1.082 \text{ km}.$$

This is the lifting condensation level, and it sits exactly in the 1–2 km range the notes quote for Earth's typical cloud base.

**$P_{\text{vap}} = 1179 \text{ Pa}$ ,  $T_d = 282.4 \text{ K}$ , and  $z_{\text{LCL}} = 1.082 \text{ km}$ , sitting in the typical 1–2 km range for Earth's cloud base.**

(b) Read the saturation-curve figure: a body with a solid surface has regions at 90 K where one atmospheric gas is present at a partial pressure of about  $10^4 \text{ Pa}$  (0.1 bar). Identify from the figure which of the five species is at its saturation point under these conditions, and deduce from the given data which body in the solar system this is. Justify your reading in one or two sentences.

**Solution.** At 90 K, only the  $\text{CH}_4$  curve passes close to  $10^4 \text{ Pa}$ : every other species has a saturation pressure many orders of magnitude below that there ( $\text{H}_2\text{O}$ ,  $\text{CO}_2$ ,  $\text{NH}_3$ ,  $\text{H}_2\text{SO}_4$  all lie far off the bottom of the plot at 90 K). The gas is methane, at its saturation point. The Lecture 6 table lists  $\text{CH}_4$  condensation on Titan, Uranus, and Neptune, but the ice giants have no solid surface at these levels: the body with a surface is Titan. The given-data table quotes 94 K for Titan's mean surface; the methane saturation pressure rises by only a factor of 1.6 from 90 to 94 K, so both temperatures sit within a factor of two of saturation for a  $\sim 0.1$  bar methane partial pressure, which is exactly why methane there cycles between vapour, rain, and lakes.

**The species is  $\text{CH}_4$  and the body is Titan: methane is at saturation near  $10^4 \text{ Pa}$  at 90 K, driving Titan's methane cycle of vapour, rain, and lakes.**

(c) The Lecture 6 notes state that Venus's sulfuric-acid cloud base is sharply defined, while Titan's methane clouds can form over a wide range of altitudes. Use the slopes of the saturation curves in the figure together with the dry lapse rates  $\Gamma_d = g/c_p$  of the two atmospheres (Venus:  $g = 8.87 \text{ m s}^{-2}$ ,  $c_p \approx 900 \text{ J kg}^{-1} \text{ K}^{-1}$ , cloud base near 350 K; Titan:  $g = 1.35 \text{ m s}^{-2}$ ,  $c_p \approx 1040 \text{ J kg}^{-1} \text{ K}^{-1}$ , clouds near 90 K) to explain both observations with the same argument.

**Solution.** Condensation begins where the local vapour pressure crosses the saturation curve, and the sharpness of a cloud base in altitude is set by how fast  $P_{\text{sat}}$  changes per kilometre of ascent:

$$\frac{d \ln P_{\text{sat}}}{dz} = \frac{L_v}{R_v T^2} \Gamma_d.$$

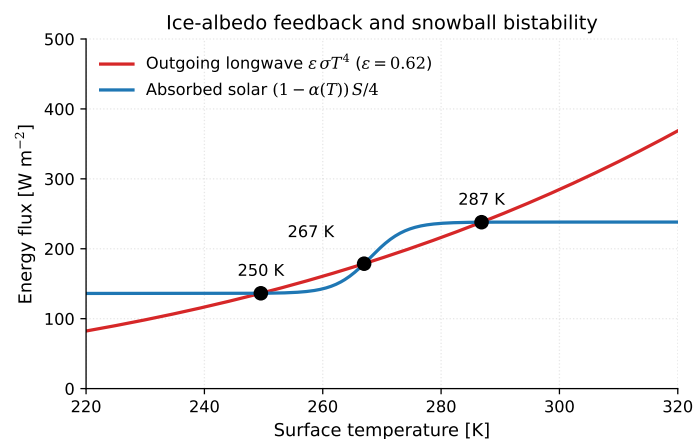
Per kelvin, the  $\text{CH}_4$  curve at 90 K is the steeper one:  $980/90^2 = 0.121 \text{ K}^{-1}$  against  $6400/350^2 = 0.052 \text{ K}^{-1}$  for  $\text{H}_2\text{SO}_4$  at 350 K, so a factor of ten in  $P_{\text{sat}}$  needs  $\ln 10/0.121 = 19 \text{ K}$  on Titan and  $\ln 10/0.052 = 44 \text{ K}$  on Venus. The lapse rates turn these temperature intervals into altitude: on Venus  $\Gamma_d = 8.87/900 \approx 9.9 \text{ K km}^{-1}$ , so the factor of ten spans only  $44/9.9 \approx 4.5 \text{ km}$  and the transition from clear air to cloud is sharp, with droplets that settle below the base evaporating within a few kilometres; on Titan  $\Gamma_d = 1.35/1040 \approx 1.3 \text{ K km}^{-1}$ , so the same factor of ten spans  $19/1.3 \approx 15 \text{ km}$ , and methane stays close to saturation over many kilometres of altitude, where clouds can appear. One equation, and it is the weak gravity of Titan, not a flat curve, that spreads its clouds.

**The saturation pressure changes with altitude at the rate  $(L_v/R_v)\Gamma_d/T^2$ : on Venus a factor of ten in  $P_{\text{sat}}$  spans about 4.5 km, so the  $\text{H}_2\text{SO}_4$  cloud base is sharp; on Titan, with  $\Gamma_d \approx 1.3 \text{ K km}^{-1}$ , it spans about 15 km, so methane clouds form across**

many kilometres of altitude.

## Problem 4 Climate on the edge

The energy-balance figure below reproduces the ice-albedo bistability diagram from the Lecture 6 notes; you will need it in part (b).



(a) The young Sun shone at 71% of its present luminosity. Compute Earth's equilibrium temperature at that time, taking today's albedo, and state the paradox this number creates when combined with the geological record.

**Solution.** The equilibrium temperature scales as the fourth root of the absorbed flux, so

$$T_{\text{eq}} = 255 \times (0.71)^{1/4} = 255 \times 0.91794 = 234.1 \text{ K.}$$

With a greenhouse like today's (+33 K), the mean surface would sit near 267 K, well below freezing, and the oceans should have been ice. Yet Hadean zircons (4.4 Ga) and Archean pillow basalts and stromatolites record liquid water and, later, life. A frozen prediction against a wet record: this is the faint young Sun paradox.

**$T_{\text{eq}} = 234.1 \text{ K}$  implies a frozen surface near 267 K (with a +33 K greenhouse), contradicting the Hadean and Archean geological evidence of liquid water and life (the faint young Sun paradox).**

(b) Read the bistability figure: identify which of the three intersections are stable and which is unstable, and describe what happens to the climate if the planet sits at the middle intersection and is nudged slightly toward warmer temperatures.

**Solution.** The outer two intersections are stable: at the 250 K crossing (snowball) and the 287 K crossing (warm state), a small temperature excursion changes emission and absorption such that the planet is pushed back toward the crossing. The middle intersection at 267 K is unstable: it sits on the steep flank of the albedo transition. Nudged warmer from 267 K, the planet melts some ice, absorbs more sunlight than it emits, and warms further; the imbalance grows until the climate lands at the stable warm state at 287 K. The same nudge cooler would run away into the snowball. One incoming sunlight curve therefore supports two possible Earths, and history decides which one you inhabit.

**The outer crossings at 250 K (snowball) and 287 K (warm state) are stable, while the middle crossing at 267 K is unstable; a warm nudge from 267 K runs away to the stable warm state at 287 K.**

**(c)** A large volcanic province doubles the volcanic CO<sub>2</sub> outgassing rate for a few million years. Using the carbonate-silicate cycle of Lecture 6, describe the chain of responses that returns Earth's climate toward balance, and explain why this thermostat no longer works on Mars.

**Solution.** The extra CO<sub>2</sub> strengthens the greenhouse and warms the surface. A warmer, wetter climate speeds up silicate weathering, both through the Arrhenius-type temperature dependence of the reactions and through increased rainfall. Faster weathering draws CO<sub>2</sub> out of the atmosphere and buries it as carbonate, so the greenhouse weakens and the temperature relaxes back over a few hundred thousand years; the new steady state is only modestly warmer, with weathering again matching the (elevated) outgassing. On Mars the loop is broken at its source: volcanism largely ceased about 3 Gyr ago, so there is no carbon resupply, and whatever CO<sub>2</sub> weathering and escape removed was never returned. A thermostat needs both a working sink and a working source; Mars kept neither.

**Elevated CO<sub>2</sub> warms Earth and speeds silicate weathering to bury carbonate until balance is restored; on Mars volcanism ceased about 3 Gyr ago, breaking the carbon source.**